

Exercises 1.4

- 5) Let $P(x)$ be the statement " x spends more than five hours every weekday in class", where the domain for x consists of all students. Express each of these quantifications in English.
- a) $\exists x P(x)$: There exists a student who spends more than five hours every weekday in class.
 - b) $\forall x P(x)$: For all students x , x spends more than five hours every weekday in class.
 - c) $\exists x \neg P(x)$: There exists a student who does not spend more than five hours every weekday in class.
 - d) $\forall x \neg P(x)$: No student spends more than five hours every weekday in class.
- 6) Let $N(x)$ be the statement " x has visited North Dakota", where the domain consists of the students in your school. Express each of these quantifications in English.
- a) $\exists x N(x)$: There exists a student in our school who has visited North Dakota.
 - b) $\forall x N(x)$: All students in our school has visited North Dakota.
 - c) $\neg \exists x N(x)$: It is not the case that there exists a student in our school who has visited North Dakota.
 - d) $\exists x \neg N(x)$: There exists a student in our school who has not visited North Dakota.
 - e) $\neg \forall x N(x)$: It is not the case that all the students in our school has visited North Dakota.
 - f) $\forall x \neg N(x)$: No students in our school has visited North Dakota.
- 7) Translate these statements into English, where $C(x)$ is " x is a comedian" and $F(x)$ is " x is funny" and the domain consists of all people.
- a) $\forall x (C(x) \rightarrow F(x))$: For the set of all people, if x is a comedian, then x is Funny, i.e. If a person is a comedian, then that person is Funny.